PathFlow: A Motion-Aware Prototyping Framework for Microscopic Pathology Video Classification

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Abstract

In recent years, AI in pathology has attracted considerable research interest; however, most existing studies focus exclusively on data collected using digital scanners. This creates a significant gap in clinical practice, as many hospitals worldwide still rely on traditional microscopy. In this paper, we address this disparity by focusing, for the first time, on AI applications for data captured directly from microscopy video feeds. We introduce MVP-CRC, the first multi-institutional dataset of 1,379 microscopy videos on colorectal carcinoma collected from seven hospitals across Vietnam with three diagnostic labels. Building on this dataset, we propose PathFlow, a motion-aware prototype-based framework that (i) selects diagnostically relevant frames using optical flow, (ii) aggregates frame-level representations via a learnable prototype bank with Top-L retrieval, and (iii) models global context with a transformer encoder for video-level prediction. Experiments on MVP-CRC show that PathFlow achieves state-of-the-art performance, reaching 91.90% in accuracy and outperforming existing approaches by 5.87%. Finally, we integrate PathFlow into an interactive web interface for real-time clinical decision support without requiring slide scanning. PathFlow is capable of reducing physician processing time by up to 50%, demonstrating a practical pathway toward deployment in resource-constrained settings. Data samples are provided at: dataverse.harvard.edu/previewurl.xhtml?token=b328291f-18d1-4911-83e4-371880d281db.

1 Introduction

Background and Motivation. Histopathological examination serves as the gold standard for cancer diagnosis [38, 40] and prognosis [44, 47]. In the traditional workflow, pathologists observe tissue slides under a microscope and manually identify pathological structures. This process is inherently labor-intensive and prone to human error due to the high cognitive load. As shown in Figure 1, the average time to screen a single slide requires a minimum of 5 minutes, and a complete case often reaches around 10 minutes. This manual burden limits diagnostic throughput and creates a bottleneck in clinical settings. More recently, the emergence of digital scanners offers a transition toward Whole Slide Imaging (WSI) [31]. However, the high cost of these scanners prevents their widespread adoption, particularly in developing nations. Our data in Figure 1 indicates that around 76% of surveyed pathologists in these regions still work exclusively with traditional microscopes. Consequently, there is an urgent need for AI-assisted systems that support automation directly on the microscope to reduce the clinical workload.

The Research Gap. AI in pathology attracts significant interest, particularly in tasks such as disease classification [35, 46], tumor segmentation [1, 37], and survival prediction [9, 42]. However, most existing studies focus on data from digital scanners [4, 23, 32]. This trend exists because whole-slide images are readily available for training. In contrast, the collection and processing of video data from

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manual microscopes is a non-trivial task. This creates a significant gap between AI research and practical clinical application in resource-limited environments (research focuses on digital scanners while microscopes are the dominator in practice). Our study addresses this gap by introducing a novel dataset and model for automated colorectal histopathology classification from microscopic videos.

Challenges. The classification of histopathology in microscopic videos faces several difficult challenges that distinguish it from standard video analysis. First, there is a severe shortage of data, as most large-scale pathology datasets focus solely on images from digital scanners [4, 12, 21, 23]. Second, microscopic videos possess unique spatio-temporal characteristics. Unlike standard videos where temporal continuity is the primary factor [2, 14], microscopic frames exhibit complex spatial relationships as the pathologist moves their view along the slide. This makes it difficult to sample the most informative frames for diagnosis. Finally, the massive size and varying resolutions of these videos lead to extreme computational complexity. Processing an entire video sequence is inefficient and often exceeds hardware limits.

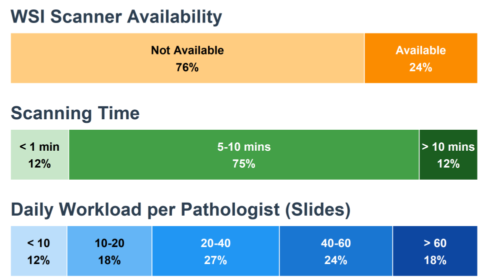


Figure 1: **Survey results across 33 hospitals in Vietnam.** Top: Availability of WSI scanners. Middle: Average scanning time per slide. Bottom: Daily diagnostic workload per pathologist.

Our Solution. To address these challenges, we develop PathFlow, a motion-aware, prototype-based framework. We curate a large-scale dataset of microscopic pathology videos from multiple hospitals, annotated by expert pathologists. PathFlow introduces an optical-flow-based frame sampling strategy [15] that accounts for spatial-temporal relationships to select diagnostically relevant frames. To mitigate computational complexity, we employ prototype learning, using a fixed set of learnable prototypes to represent diverse inputs, thereby maintaining efficiency across varying video lengths and resolutions.

Our Contributions. Our primary contributions are summarized as follows:

- We construct the first multi-institutional, large-scale microscopic video dataset for colorectal carcinoma, featuring 1,379 videos from seven hospitals with three diagnostic labels for classification tasks (Section 3).
- We propose PathFlow, a novel model that automates colorectal pathology classification. Our model outperforms current methods by a minimum of 5.87% in accuracy. Furthermore, we pilot-test and evaluate the model at a major hospital, where it achieves an accuracy of 92% and reduces diagnostic time by nearly 50% compared to manual procedures (Section 4).
- To address the inherent challenges of processing pathology videos, we introduce two novel techniques. First, we propose a motion-aware frame sampling method based on optical flow, designed to capture the unique scanning dynamics of microscopy. This approach ensures the selection of the most diagnostically significant frames. Second, we implement a prototype banking mechanism that effectively mitigates the high computational costs associated with high-resolution medical video analysis (Section 4).

2 Related Work

2.1 Whole Slide Image Classification

Most computational pathology frameworks for histopathological classification operate on whole slide images (WSIs) and formulate prediction as a multiple instance learning (MIL) problem [11, 30], in which a slide is represented as an unordered bag of image patches under weak supervision. Early attention-based MIL methods [17, 27, 33] demonstrate the effectiveness of instance-level importance weighting, and subsequent work further refines this paradigm to improve robustness and interpretability by addressing unreliable attention, spurious correlations, and multimodal integration through attention masking [43], multimodal MIL formulations [35], causal learning objectives [8], and continual learning [45].

More recently, structured and sequential representations of WSIs have been explored to capture contextual dependencies beyond unordered aggregation. Transformer-based models, memory-augmented architectures [3], and video-structured MIL approaches [29] reinterpret WSIs as ordered patch sequences or pseudo-videos to enable long-range reasoning. However, these methods still rely on scanner-produced WSIs and impose artificial ordering on static content, limiting their ability to model temporal dynamics and acquisition variability. As a result, they are ill-suited for microscope-acquired pathology videos with variable length and motion-induced redundancy.

2.2 Video Datasets for Whole Slide Analysis

The progress of learning-based methods in computational pathology is closely driven by the availability of curated datasets. Most existing resources [4, 23] focus on static histopathological data, particularly scanner-generated WSIs or collections of image patches derived from individual patient cases. These datasets support substantial advances in WSI classification, segmentation, and representation learning, as well as the training and benchmarking of large-scale foundation models [22, 29, 41, 45], and continue to serve as standard reference benchmarks within the field.

In contrast, pathology video datasets remain scarce and are largely confined to adjacent domains such as endoscopy [21] and colonoscopy [12], which capture macroscopic anatomy rather than cellular-level morphology. Consequently, large-scale microscope-acquired histopathology video datasets with temporal continuity and real-world acquisition variability are limited, hindering the development of learning frameworks aligned with realistic clinical workflows and motivating a shift toward microscopy-based pathology.

2.3 Motion Modeling with Optical Flow in Pathology

Optical flow [15] is a fundamental representation of motion in visual perception widely used in computer vision for temporal correspondence, motion estimation, and long-form video understanding [14, 39]. Recent studies highlight its role as an effective motion prior [25] for guiding feature aggregation and reasoning in natural video settings.

In computational pathology, optical flow is predominantly used for geometric preprocessing, while its potential for diagnostic modeling remains underexplored. Existing work mainly exploits optical flow for frame alignment and stitching, particularly for reconstructing whole-slide images from microscopic videos [20]. Its use as a diagnostic signal (e.g., identifying informative transitions, reducing temporal redundancy, or guiding learning from pathology videos) remains limited. In contrast to natural video understanding, this gap motivates motion-aware pathology frameworks that explicitly integrate optical flow into learning from microscope-acquired data.

3 Our Proposed MVP-CRC Dataset

Our dataset, MVP-CRC (**M**icroscopic **V**ideos in **P**athology for **C**olo**R**ectal **C**arcinoma), consists of 1,379 microscopic pathology video studies collected from multiple hospitals nationwide. Each study corresponds to a single case of colorectal biopsy and contains one video per case. In this section, we detail the data curation process and provide MVP-CRC's statistics.

3.1 Data Curation Process

The data curation process is illustrated in Figure 2. Each video is captured directly from an optical microscope using a microscope-mounted camera during routine histopathological examination. During acquisition, the camera records a limited field of view over localized tissue regions, which is systematically navigated across the slide to ensure coverage of diagnostically relevant areas. Video acquisition terminates once all non-background tissue regions are visually examined, reflecting standard clinical workflows in settings where WSI scanners are limited or unavailable.

All videos are recorded under bright-field microscopy at a fixed magnification of 20×, with no magnification changes during recording. In real-world settings, the most-used magnification level for diagnosis inspection is 20×, which is why our pathologists chose to record videos for MVP-CRC with the same magnification level to best reflect real clinical practice. Tissue sections are prepared using hematoxylin and eosin (H&E) staining according to standard pathology protocols. Videos capture pathologists' manual navigation across regions of interest (ROIs), producing temporally ordered sequences characterized by continuous motion. Each case is assigned a video-level diagnostic label:

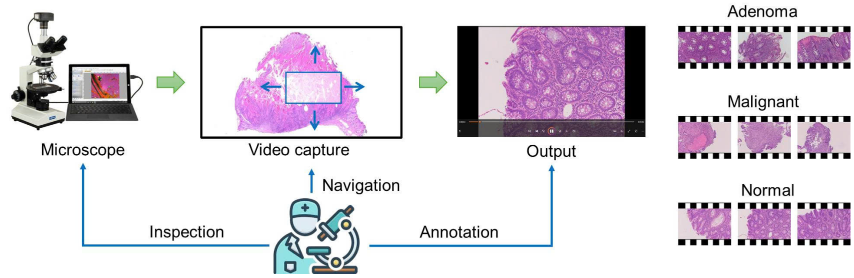


Figure 2: **Microscopic video acquisition workflow.** To capture each video, pathologists navigate tissue slides under an optical microscope while recording via a microscope-attached camera. Example frames from each class (Adenoma, Malignant, and Normal) cases are shown.

Adenoma (A), *Malignant (M)*, or *Normal (N)*, by board-certified pathologists with over 10 years of clinical experience. Labels are verified through inter-observer review: one doctor performs manual inspection while another views the microscope optic. Both agree on the video's label before inclusion in the dataset. All videos are anonymized to protect patient privacy.

3.2 Dataset Statistics

The distribution of labels is shown in Table 1. Videos are recorded with a variety of microscope brands, including Olympus, Leica, and Axiolab. The dataset is collected from seven hospitals with a total of 43 hours of footage. The most common spatial resolutions are 1920×1440 and 1600×1200, though variations exist depending on the camera-microscope interface of each local hospital. Video length ranges from 2 to 787 seconds with an average of 120 seconds. Frame count ranges from 25 to 37,796 frames with an average of 1,967 frames. Detailed statistical analysis about the duration and number of frames per video is provided in Appendix A.

Table 1: **Label distribution by acquisition devices.**

Acquisition device	Adenoma	Malignant	Normal	Total
Olympus BX53	311	297	227	835
BX43	69	62	59	190
CX43	29	15	31	75
CX41	47	20	28	95
Leica DM1000	30	40	14	84
Axiolab 5	40	30	30	100
Total	**526**	**464**	**389**	**1379**

4 The PathFlow Architecture

While disease classification from microscopic videos can be framed as a video classification task, standard methods struggle with the unique characteristics of pathology sequences. Specifically, microscopic videos present two primary challenges: (1) Information density vs. Signal sparsity, where massive data volumes are coupled with diagnostic markers localized in sparse regions; and (2) Complex spatial-temporal coupling, which requires modeling intricate dependencies to capture the clinician's diagnostic scanning patterns.

To address these challenges, PathFlow integrates three key components. First, a Motion-Aware Frame Selection algorithm (Sec. 4.1) employs optical-flow analysis to remove redundant frames and retain diagnostically relevant observations. Second, a Global Prototype Aggregation mechanism (Sec. 4.2) compresses long sequences into a compact set of representative tokens while preserving critical morphological features. Finally, a Relational Context Modeling module (Sec. 4.3) utilizes a Transformer-based architecture to capture global spatial and temporal dependencies among prototypes for accurate diagnostic prediction. Figure 3 presents an overview of PathFlow.

4.1 Feature Extraction

Optical Flow-Based Frame Selection. Unlike conventional videos, microscopic pathology sequences exhibit tightly coupled spatial-temporal dynamics, making standard frame sampling methods [13, 14, 39] inadequate for diagnostic use. We therefore propose a motion-aware, optical-flow-based sampling strategy that prioritizes frames associated with deliberate scanning or focused

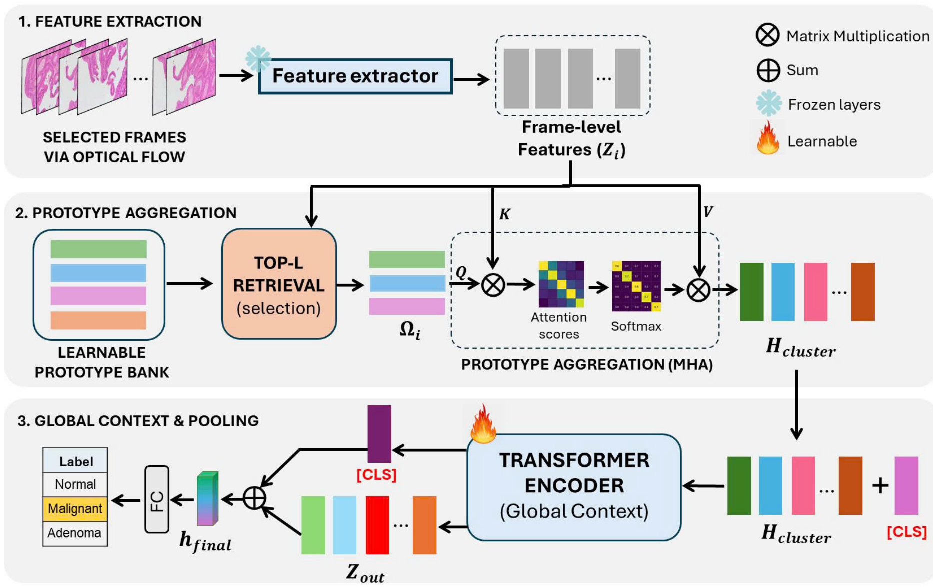


Figure 3: **Overview of our proposed PathFlow framework for microscopic pathology video classification.** Selected video frames are encoded with a feature extractor before being used to aggregate a learnable prototype bank. The aggregation output is modeled using a transformer encoder for video-level classification.

inspection such as raster scanning, while suppressing redundant or low-quality frames caused by blur, jitter, or insufficient tissue content. The overall design is described in Algorithm 1, following three core principles:

(1) Robust global motion estimation: Given a sequence of frames $F = \{f_t\}_{t=1}^{T}$, we extract a central crop from each frame to mitigate boundary artifacts. Within this region, up to D_p sparse Shi-Tomasi [34] keypoints are detected and tracked between consecutive frames using Lucas-Kanade [28] optical flow, where D_p denotes the maximum number of tracked features. Let v_u be the displacement vector of the u-th tracked keypoint, with $u \in \{1, \dots, D_p\}$. To obtain a stable estimate of camera motion resilient to local outliers like cellular deformation or debris, we define the dominant global displacement as $\Delta_t = \mathrm{median}_u(v_u)$. The corresponding motion magnitude is given by $s_t = \|\Delta_t\|$.

(2) Motion quality filtering: Each candidate frame undergoes a validity check to ensure reliable motion estimation. Frames are discarded if the number of detected keypoints falls below a threshold T_{feat}. To suppress motion blur and camera shake, we compute a motion inconsistency measure $J_t = \mathrm{median}_u(\|v_u - \Delta_t\|)$, which captures the dispersion of local flows around the dominant motion. Frames with negligible displacement ($s_t < T_{\text{motion}}$) or excessive jitter ($J_t > T_{\text{jitter}}$) are rejected to improve robustness against blur and deformation caused by magnification changes. To adapt to varying scanning speeds, a rolling history buffer H stores the most recent N_{hist} motion magnitudes, and a robust reference speed is estimated as $\mu = \mathrm{median}(H)$.

(3) Adaptive distance- and direction-based triggering: For valid frames, dominant motion vectors are accumulated into a global displacement A, representing camera traversal since the last selection. The algorithm distinguishes rapid raster scanning from slower inspection by comparing the instantaneous speed s_t with a historical reference μ. When a speed drop is detected ($s_t < \tau\mu$), the algorithm enters inspection mode and reduces the sampling distance using a scaling factor $\alpha = \alpha_{\text{insp}} < 1$; otherwise, $\alpha = 1.0$. A new frame is selected when $|A| \geq \alpha W(1-\lambda)$, where W is the frame width and λ controls overlap, or when a sharp direction change is detected. The selected frames are denoted by $\mathbf{X}$.

UNI-Based Feature Extraction. The selected frames are subsequently processed by a feature extractor. Specifically, we utilize the encoder of the UNI foundation model [10] pretrained on an extensive corpus of pathology images to ensure robust feature representation. For every frame $\mathbf{x}_{i,t}$ in the input sequence $\mathbf{X}_i \in \mathbb{R}^{T_i \times H \times W \times 3}$, we extract a D-dimensional feature embedding $\mathbf{z}_{i,t} \in \mathbb{R}^D$:

$$\mathbf{z}_{i,t} = \mathrm{UNI}(\mathbf{x}_{i,t}), \tag{1}$$

where D is set to 1536 following the setting of the UNI2 backbone configuration [10]. Consequently, each video $\mathbf{X}_i$ is transformed into a feature tensor $\mathbf{Z}_i = \{\mathbf{z}_{i,t}\}_{t=1}^{T_i}$ with the corresponding ordering of frames as the original video. These tensors are pre-computed and stored, forming the training dataset $\mathscr{D} = \{(\mathbf{Z}_i, y_i)\}_{i=1}^{N}$. This approach allows the downstream model to focus exclusively on learning inter-frame dynamics and morphological interactions without having to repeatedly extract features.

4.2 Prototype Aggregation

Directly applying standard Transformers to raw frame sequences $\mathbf{Z}_i$ is computationally prohibitive due to their quadratic complexity, $O(T_i^2)$, and the overlap of temporally adjacent frames. To address this, PathFlow employs a hierarchical aggregation mechanism that compresses long, redundant sequences into a compact set of diagnostically salient tokens using a learnable bank of global prototypes. While prototype-based representation learning has been explored in related contexts [24, 35, 36], our approach differs in its fundamental architecture. Existing methods typically construct sample-specific prototype banks to capture morphological patterns within individual videos. In contrast, our method learns a global, shared prototype bank across the entire dataset. This design allows the model to capture recurring disease-specific features while maintaining sensitivity to intra-sample variability.

Learnable Prototype-Based Aggregation. We maintain a learnable prototype bank $\mathscr{P} = \{\mu_k\}_{k=1}^{K}$, where K denotes the maximum number of prototypes and $\rho \in (0,1]$ controls the retrieval budget relative to sequence length. Each prototype encodes a latent histological concept shared across the training distribution. Given frame-level features $\mathbf{Z}_i = \{\mathbf{z}_{i,t}\}_{t=1}^{T_i}$, a subset of prototypes is retrieved as:

$$L_i = \lfloor \rho \cdot T_i \rfloor. \tag{2}$$

Prototype Retrieval. We first compute the cosine similarity between the normalized frame features $\mathbf{z}_{i,t}$ and the global prototypes μ_k:

$$s_{t,k} = \frac{\mathbf{z}_{i,t}^{\top} \mu_k}{\|\mathbf{z}_{i,t}\| \|\mu_k\|}. \tag{3}$$

Then, for each frame t, we determine the subset S of L_i prototypes in $\mathscr{P}$ that are most similar to $\mathbf{z}_{i,t}$. We denote this set as:

$$\mathrm{TopL}_{L_i}^{\mathrm{sim}}(\mathbf{z}_{i,t}, \mathscr{P}) = \underset{S \subseteq \mathscr{P}, |S| = L_i}{\arg\max} \sum_{\mu_k \in S} s(\mathbf{z}_{i,t}, \mu_k). \tag{4}$$

Each frame-level feature $\mathbf{z}_{i,t}$ has a corresponding subset of Top-L_i prototypes. In other words, the sequence of Top-L_i subsets still follows the order of the original sequence of frame-level features $\mathbf{Z}_i$. Given these retrieval results, a per-video frequency score is computed for each prototype μ_k:

$$F_i[\mu_k] = \sum_{t=1}^{T_i} \mathbb{I}(\mu_k \in \mathrm{TopL}_{L_i}^{\mathrm{sim}}(\mathbf{z}_{i,t}, \mathscr{P})). \tag{5}$$

Intuitively, the frequency score $F_i[\mu_k]$ represents the total number of frames in video i that selected prototype μ_k as one of their top matches. Finally, we determine the final video-level set of top L_i prototypes, denoted as Ω_i, by selecting those with the highest frequency scores:

$$\Omega_i = \underset{I \subseteq \{1,\dots,K\}, |I| = L_i}{\arg\max} \sum_{k \in I} F_i[\mu_k]. \tag{6}$$

Cross-Attention-Based Refinement. Following the retrieval of Ω_i, we refine these prototypes using a Multi-Head Attention (MHA) module conditioned on the local temporal context. As shown in Figure 3, the retrieved prototypes Ω_i act as the queries, while the frame-level feature sequence $\mathbf{Z}_i$ provides the keys and values, enabling each prototype to selectively aggregate relevant frames into representative tokens. The refinement process is defined as:

$$\mathbf{H}_{\mathrm{cluster}} = \mathrm{Softmax}\left(\frac{\Omega_i \mathbf{Z}_i^{\top}}{\sqrt{d_k}}\right) \mathbf{Z}_i, \tag{7}$$

where $\mathbf{H}_{\text{cluster}} \in \mathbb{R}^{L_i \times D}$ represents a set of compressed, representative tokens. This bottleneck architecture ensures that subsequent layers only process L_i tokens ($L_i \ll T_i$), significantly reducing the computational overhead while preserving critical diagnostic information.

4.3 Global Context

The aggregated prototypes $\mathbf{H}_{\text{cluster}}$ distill the redundant frame sequence into a compact set of salient morphological concepts. To capture the complex relationships and co-occurrences among these diverse tissue features, we process $\mathbf{H}_{\text{cluster}}$ through a Transformer Encoder. We prepend a learnable classification token, $x_{[CLS]}$, to the sequence:

$$\mathbf{Z}_{in} = [x_{[CLS]} \parallel \mathbf{H}_{\text{cluster}}]. \tag{8}$$

This sequence is processed by Transformer layers to produce the contextualized output $\mathbf{Z}_{out}$, capturing dependencies between histological patterns observed during the scan.

Hybrid Feature Pooling. Relying solely on the $[CLS]$ token can sometimes lead to over-specialization to dominant features [1]. To mitigate this, we employ a Hybrid Pooling strategy in the final classification head. We combine the global context of the $[CLS]$ token with the distributed evidence from the cluster tokens:

$$\mathbf{h}_{\text{final}} = \alpha \cdot \mathbf{Z}_{\text{out}}^{[CLS]} + (1-\alpha) \cdot \frac{1}{K} \sum_{k=1}^{K} \mathbf{Z}_{\text{out}}^{k}, \tag{9}$$

where α is a hyperparameter. This formulation balances the diagnostic decision of the $[CLS]$ token with aggregated prototyping evidence, ensuring that the model accounts for the entire slide content.

Objective Function. The final representation $\mathbf{h}_{\text{final}}$ is passed through a linear projection head to generate the prediction logits $\hat{\mathbf{y}}_i$. The model parameters are optimized end-to-end by minimizing the standard Cross-Entropy loss over the training dataset:

$$\mathscr{L}(\theta) = -\frac{1}{N} \sum_{i=1}^{N} \sum_{c=1}^{C} \mathbb{I}(y_i = c) \log(\hat{y}_{i,c}). \tag{10}$$

This objective ensures that the learned cluster tokens adaptively align with the most discriminative regions of the video, effectively recovering underlying diagnostic signals from noisy panning data.

5 Experiments and Results

5.1 Experimental Settings

Baselines. To ensure comprehensive benchmarking, we compare our proposed framework against a diverse set of established state-of-the-art methods on the MVP-CRC dataset. These include Multiple Instance Learning (MIL) strategies such as ABMIL [17], ACMIL [43], AttriMIL [6], and CLAM [27]. Additionally, we validate our method against traditional video analysis approaches, specifically TimeSFormer [5] and Swin Transformer variants [26]. Finally, we present ablation studies in Section 5.3. All implementation details are provided in Appendix B.

Evaluation Metrics. We assess the performance of the classification framework using three standard metrics: Accuracy, F1-Score, and Area Under the Receiver Operating Characteristic (AUC-ROC). Given the medical context of the task, where the cost of missing a malignant case is high, we place particular emphasis these metrics to ensure the model maintains high discriminative power and robustness in identifying pathological features.

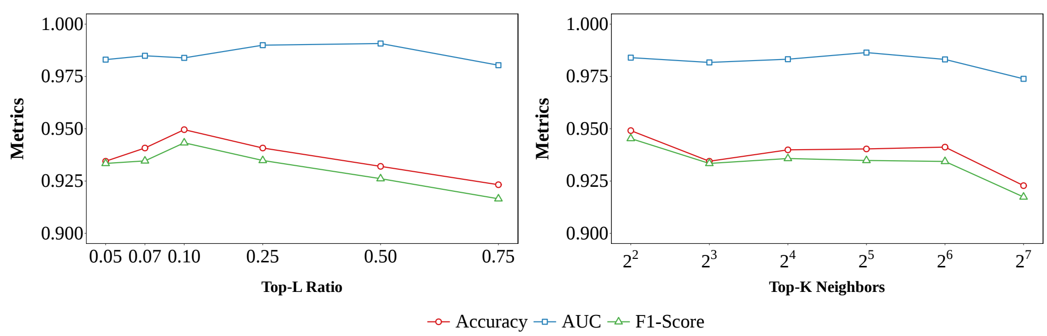
5.2 Comparison with State-of-the-Art Methods

Table 2 reports a quantitative comparison between PathFlow and state-of-the-art MIL- and video-based baselines over five independent runs. We report both average (Top-5) and best-run (Top-1)

[1] We further support this statement in Appendix F

Table 2: **Comparison of PathFlow against state-of-the-art MIL-based and video-based baselines.** Results are reported under Top-1 (best run) and Top-5 (average over all runs) evaluation. **Bold** and underline indicate the best and second-best performing approach, respectively. The *Improvement* row reports the absolute performance gain of PathFlow over the strongest competing method.

Methods	Top-1			Top-5		
	ACC (%)	F_1-Score (%)	AUC (%)	ACC (%)	F_1-Score (%)	AUC (%)
ABMIL	<u>86.91</u>	<u>86.91</u>	96.61	<u>86.03 ± 0.88</u>	<u>86.59 ± 0.47</u>	96.56 ± 0.20
ACMIL	84.52	84.52	<u>96.85</u>	85.00 ± 1.21	85.00 ± 1.21	96.37 ± 0.42
AttriMIL	57.94	57.07	80.52	59.04 ± 0.95	57.95 ± 0.97	78.78 ± 0.90
CLAM	84.52	84.89	95.94	85.24 ± 1.21	85.32 ± 1.16	<u>96.62 ± 0.40</u>
TimeSFormer	84.92	84.74	95.99	84.13 ± 0.65	83.90 ± 0.65	95.36 ± 0.47
Swin-B	83.33	83.08	94.49	82.54 ± 0.65	82.19 ± 0.72	94.78 ± 0.24
Swin-S	82.54	81.85	92.97	79.36 ± 2.25	79.06 ± 1.97	92.91 ± 0.26
Swin-T	84.92	84.45	94.78	83.86 ± 1.50	83.39 ± 1.70	95.20 ± 0.39
PathFlow	**90.61**	**90.33**	**96.93**	**91.90 ± 1.25**	**91.52 ± 1.19**	**96.78 ± 0.45**
Improvement	*+3.70*	*+3.42*	*+0.08*	*+5.87*	*+4.93*	*+0.16*



(a) Performance vs. Top-L Ratio (b) Performance vs. Top-K Neighbors

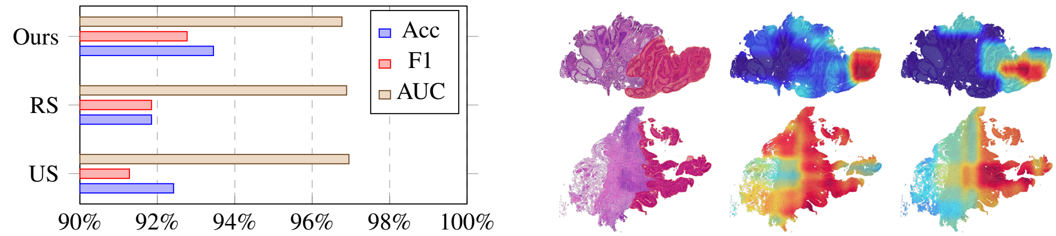
Figure 4: **Ablation studies on the number of selected prototypes in PathFlow.** (a) Top-L ratio and (b) Top-K neighbors selection on validation performance.

performance. Under Top-5 evaluation, PathFlow achieves the best results across all metrics, with 91.90% Accuracy, 91.52% F1-score, and 96.78% AUC, outperforming the strongest baseline by +5.87%, +4.93%, and +0.16%, respectively. PathFlow also attains the highest Top-1 Accuracy (90.61%), F1-score (90.33%), and AUC (96.93%), demonstrating consistent performance gains under stricter evaluation. Overall, these results validate the effectiveness of our proposed motion-aware, prototype-based design for microscopic pathology video classification.

5.3 Ablation Studies

Effect of the Number of Selected Prototypes. We analyze the sensitivity of PathFlow to the number of selected prototypes under adaptive and fixed retrieval strategies. In the adaptive setting, a fixed prototype bank is maintained while a subset is dynamically retrieved according to a retrieval ratio r, defined as the percentage of selected prototypes. We evaluate $r \in \{5\%, 7\%, 10\%, 25\%, 50\%, 75\%\}$. As shown in Figure 4, performance peaks at $r = 10\%$, suggesting the best balance between representational diversity and retrieval noise.

For the fixed strategy, we vary the prototype bank size (Top-K) with $K \in \{2^2, 2^3, 2^4, 2^5, 2^6, 2^7\}$, corresponding to between 4 and 128 prototypes. A fixed number of nearest prototypes is retrieved per query without adaptive selection. As shown in Figure 4, the best performance is achieved at $K = 2^2$, suggesting that a compact prototype bank is sufficient to capture discriminative patterns in the proposed dataset.



(a) Quantitative Evaluation (b) GT (c) US (d) Ours

Figure 5: **Quantitative and qualitative comparison of frame selection strategies.** (a) Quantitative comparison against uniform sampling (US) and random sampling (RS). (b) Ground-truth (GT) annotation. (c) Uniform sampling (US) produces diffuse attention over healthy and neoplastic regions. (d) Our method selectively focuses on diagnostically relevant regions.

Impact of Optical Flow-Based Frame Selection. To evaluate the effectiveness of our frame selection strategy, we compare it against standard baselines both quantitatively and qualitatively. As shown in Figure 5a, our method achieves an accuracy of 93.45%, outperforming Uniform Sampling (US) at 92.42% and Random Sampling (RS) at 91.85%, demonstrating that optical-flow-guided selection provides more informative frames than arbitrary sampling strategies. Figures 5b-5d further compare saliency distributions with ground-truth annotations from two exemplar cases in the MVP-CRC dataset. Uniform Sampling produces diffuse attention over healthy and neoplastic regions, whereas our method more accurately localizes diagnostically relevant areas. These results indicate that our proposed strategy better focuses on class-indicative regions.

Comparison of Feature Extraction Backbones. To evaluate the robustness of our framework and the impact of visual representation quality, we conduct an ablation study comparing various pretrained feature extractors. We evaluate two lightweight backbones, ViT-S/16 pretrained using DINO [7] on a collection of 36,666 WSIs [18] and ResNet-50 [16] alongside two larger backbones, UNI [10] and Virchow2 [48]. As shown in Table 3, UNI shows the strongest performance while being significantly more stable than the other backbones. The consistent performance gains observed with UNI justify its selection as the default feature extractor for the duration of this study.

Table 3: **Performance of different feature extraction backbones. Bold** and underline indicate the backbone with the best and second-best performance, respectively.

Backbone	ACC (%)	F_1-Score (%)	AUC (%)
UNI	**89.41 $\pm$ 0.34**	**89.04 $\pm$ 0.30**	**96.65 $\pm$ 0.29**
Lunit	88.29 $\pm$ 1.85	88.11 $\pm$ 1.68	96.29 $\pm$ 0.78
Virchow2	89.38 $\pm$ 2.85	88.85 $\pm$ 2.79	96.48 $\pm$ 1.32
ResNet-50	77.81 $\pm$ 2.94	77.72 $\pm$ 2.84	89.92 $\pm$ 3.03

6 Conclusion

In this work, we studied colorectal cancer histopathology classification using microscope-captured videos rather than scanner-produced WSIs. We introduced MVP-CRC, a multi-institutional dataset of 1,379 microscopic videos from seven hospitals annotated by expert pathologists, and proposed PathFlow, a motion-aware, prototype-based framework integrating optical-flow-guided frame selection, prototype aggregation, and global context modeling to assist pathologists in clinical decision-making. PathFlow consistently outperformed representative MIL-based and video-based baselines under evaluations, while enabling practical deployment through a lightweight web-based interface compatible with scanner-limited clinical workflows. Future directions include improving PathFlow's robustness across diverse acquisition conditions, extending the framework to additional diagnostic tasks like ROI segmentation, and expanding MVP-CRC by providing detailed frame-level annotations of label-indicative ROIs for polyp subtyping to further its clinical utility.

Software and Data

The source code for PathFlow is available at: anonymous.4open.science/r/PathFlow-5B84. A subset of the MVP-CRC dataset is provided at: dataverse.harvard.edu/previewurl.xhtml?token=b328291f-18d1-4911-83e4-371880d281db.

Upon acceptance, we will publicly release the dataset in compliance with the collaborating hospitals' ethical and data governance regulations. Researchers who wish to access the full dataset will be granted access after completing the required procedures and approvals specified by the hospitals.

Impact Statement

This paper presents work whose goal is to advance the field of machine learning for computational pathology by providing a novel dataset for analysis of microscope-acquired histopathology videos without reliance on whole-slide scanners. The potential societal impact of this work lies in improving the accessibility of AI-assisted diagnostic tools in resource-constrained clinical settings.

Ethical Statement

The data utilized in this research were obtained with the formal approval and informed consent of the participating hospitals. To ensure patient privacy and confidentiality, all data have been fully de-identified and anonymized, with all personally identifiable information removed in accordance with institutional ethical guidelines and data protection regulations.

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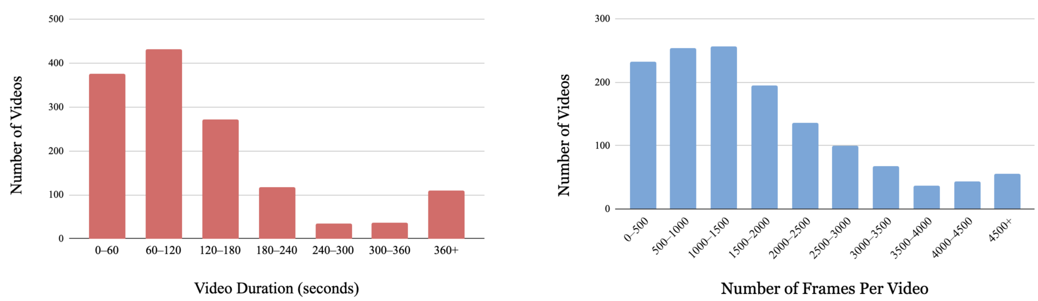
A Dataset Statistics

The MVP-CRC dataset exhibits significant diversity in both temporal duration and frame density, which highlights the need for models that generalize well across varying clinical protocols. Figure 6 provides a detailed breakdown of these distributions based on the 1,379 microscopic pathology video studies collected.

Video Duration Analysis. As illustrated in Figure 6a, the distribution of video duration is positively skewed with a significant concentration of studies in the short-to-medium range. In many clinical biopsy screenings, they are captured for approximately one to three minutes of active scanning. Meanwhile, over 100 videos are more than six minutes long. This variance, ranging from 2 to 787 seconds for each video, highlights the necessity for efficient temporal indexing to maintain a consistent computational budget during inference regardless of the total video length.

Frame Count Distribution. The distribution of frame counts, shown in Figure 6b, mirrors temporal trends while offering insight into the data density available for feature extraction. The majority of the dataset contains 1,500 or fewer frames, aligning with the standard frame sampling rates of the acquisition devices used across the seven participating hospitals. Frame counts can reach a maximum of 37,796 frames due to different video capture configurations. This distribution justifies the use of adaptive frame sampling strategies to extract salient diagnostic features from high-frame-count studies without overburdening computational overhead.



(a) Video duration

(b) Frame count

Figure 6: **Distribution of video duration and frame count per video in the MVP-CRC dataset.** (a) Statistics on the duration of video samples. (b) Statistics on the number of frames of video samples.

B Implementation Details

We implement PathFlow as detailed above. The training, test and validation sets are formed with a $70:15:15$ ratio comparative to the dataset we use for implementation, MVP-CRC. For the motion-aware frame selection module, we employ the following optical flow hyperparameters: a minimum feature count $T_{\text{feat}} = 10$, a minimum motion threshold $T_{\text{motion}} = 2$ pixels, a maximum jitter tolerance $T_{\text{jitter}} = 3$ pixels, a cosine turn threshold $T_{\text{turn}} = 0.3$, and a target overlap ratio $\lambda = 0.4$. We utilize the UNI [10] architecture as our backbone, featuring a hidden dimension of size 1536 and a patch size of 16. The number of selected prototypes is defined by an adaptive retrieval strategy, with a maximum prototype bank size of 1024 and a selection ratio r =10%. Training is performed using the Adam [19] optimizer with an initial learning rate of 2×10^{-4} and a weight decay of 1×10^{-4}. All experiments are seeded (1026) for reproducibility and run on an NVIDIA RTX 4090 GPU. The training process requires approximately 20GB of VRAM, with each training session lasting around 2 to 3 days. We use a batch size of 8 and set a maximum of 100 training epochs with an early stopping patience of 30 epochs.

C Demonstration

To assess the operational efficiency of the PathFlow framework, we develop a streamlined web interface tailored for clinical environments, shown in Figure 7. This tool minimizes the manual overhead of diagnostic screening by providing an automated workflow from raw video data to final classification. This section describes the system behavior and the rapid turnaround achieved during real-time inference. We provide a demonstration video for this interface at: anonymous.4open.science/r/PathFlow-5B84

Workflow and Clinical Utility. The PathFlow interface prioritizes functional minimalism to maximize time savings for physicians. By eliminating unnecessary visualization layers, the system allows clinicians to focus exclusively on the final diagnostic output. The workflow involves only the submission of a video sample, after which PathFlow executes automated spatial normalization and temporal indexing. This process ensures that complex data is optimized for the transformer-based backbone without requiring manual technical configuration from the medical staff.

Diagnostic Output and Decision Support. For each processed sample, PathFlow generates a categorical classification representing Normal, Adenoma, or Malignant states, accompanied by a softmax probability score. This confidence metric provides physicians with immediate decision support based on the model's temporal feature aggregation. The interface delivers high-speed inference results, which is essential for maintaining clinical throughput in high-pressure medical settings where rapid decision-making is critical.

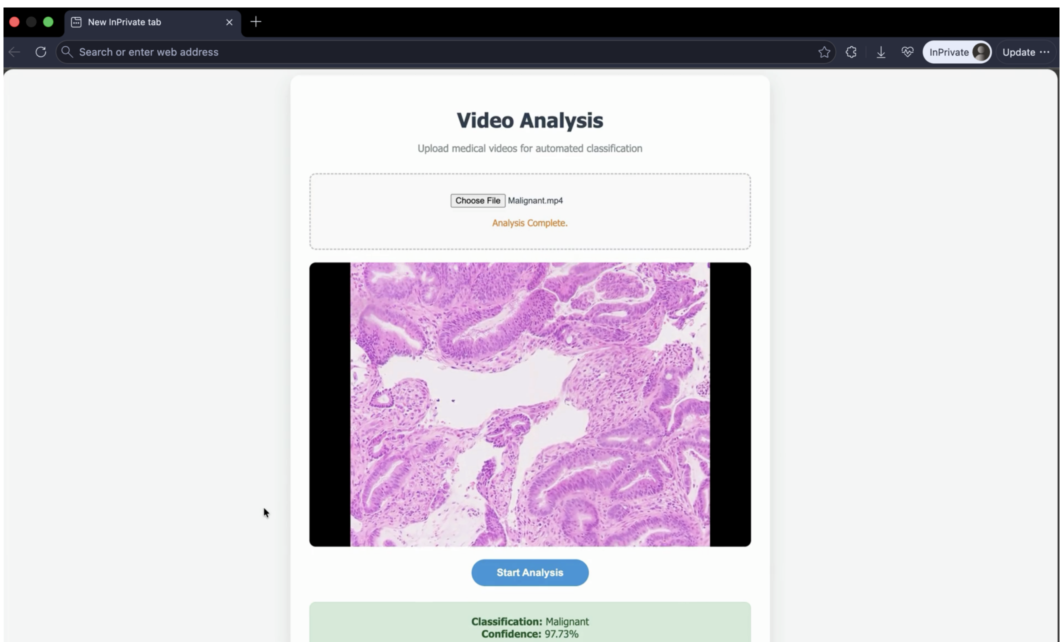


Figure 7: **Demonstration of the PathFlow clinical web application user interface.** Automated analysis of an uploaded histopathology video is provided alongside its predicted diagnostic classification and confidence score. The interface is simply designed for transparency and user-friendliness with clinicians.

D Road to Deployment

Our deployment strategy emphasizes close collaboration with practicing pathologists and alignment with existing clinical workflows. PathFlow is implemented as a lightweight web-based decision-support tool that allows pathologists to upload microscope-acquired videos and receive diagnostic suggestions without requiring slide scanners or changes to acquisition protocols. Deployment at a major hospital in Vietnam achieves approximately 92% accuracy with an average processing time of 2-5 minutes per case, resulting in an estimated 50% reduction in review time compared to scanner-based workflows, often taking 5-10 minutes per case. This time includes both the model's inference and system latency factors.

The interface[2], demonstrated in Appendix C, is broadly accessible and requires only internet connectivity, enabling deployment across diverse clinical environments. Feedback from pathologists has highlighted its value in supporting junior practitioners with rapid, consistent feedback and reducing workload for experienced clinicians through automated classification, while mitigating clinician oversight. Joint evaluation with clinical collaborators indicates that an accuracy of approximately 90% is sufficient for early-stage diagnostic support; our current results meet this requirement, with ongoing work focused on improving robustness and generalization through continued clinical feedback.

E Additional Experiment Results

In clinical practice, histopathology videos are routinely captured using a multitude of hardware configurations across different hospitals, introducing significant variations in illumination, color schemes, and sensor noise primarily due to differences between microscope brands. To evaluate the intra-domain robustness of PathFlow across these varied acquisition settings, we run performance tests by microscope source. For each subset of data attained from a distinct acquisition setting, we perform an independent 70 : 30 train-test split and train a dedicated submodel exclusively on data from that specific microscope before evaluating its performance on the corresponding 30% holdout set.

Table 4: **Performance evaluation of PathFlow in various acquisition settings.** Results are reported in Accuracy, F1-Score, and AUC. Overall, PathFlow is fairly robust in terms of prediction capabilities in different acquisition contexts.

Microscope Setup	Accuracy	AUC	F1 Score
Olympus BX53	0.9061	0.9033	0.9693
Olympus BX43	0.8614	0.9422	0.8601
Olympus CX43	0.7111	0.8534	0.7125
Leica DM1000	0.8910	0.9850	0.9092
Axiolab 5	0.8889	0.9436	0.8885

Table 4 presents the performance of PathFlow across different microscope acquisition setups. Overall, the model demonstrates stable and competitive performance under varying imaging conditions, achieving consistently strong metrics across most configurations. These results suggest that the proposed motion-aware sampling and prototype aggregation framework remains effective despite vast changes in microscope hardware, illumination, and acquisition characteristics commonly encountered in real-world clinical environments.

F Sensitivity of α in Hybrid Feature Pooling

We evaluate the effect of changing the value of α in the hybrid feature pooling module to show that relying completely on the $[CLS]$ token does not always guarantee best results. As shown in Figure 8, performance peaks at an α value of 0.8, indicating that relying fully on the $[CLS]$ token is not optimal. We speculate that varying the value of α downplays the role of the $[CLS]$ token while also allowing the model to consider the cluster tokens learned along the training process. Thus, we set α to 0.8 for all further experiments.

[2]We also provide a demonstration video in our repository

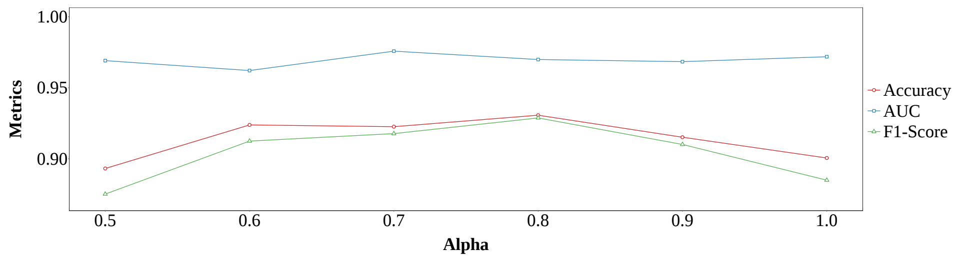


Figure 8: **Comparison of different α values in the hybrid feature pooling module.**

G Ablation Studies on Our Optical-Flow-based Frame Selection Strategy

Filtering out noisy frames. A notable problem with microscope-view videos is that changes in magnification levels or high-speed motion may introduce severe focal blur or deformation. Our proposed Algorithm 1 intrinsically mitigates this through a pause-and-recover mechanism:

- **Filter Rejection:** Blurry frames fail either the feature minimum ($|P_{prev}| < T_{feat}$) or jitter limit ($J_t > T_{jitter}$) and are skipped.
- **Tracking Suspension:** The anchor index ($prev$) and spatial accumulator (A) only update on valid frames, pausing tracking until focus returns.

While pausing risks overlap loss during simultaneous X-Y panning, in real clinical practice, panning is typically halted during Z-axis adjustments. To further study quality of this filter mechanism, we conduct an ablation study across a subset of 835 videos. Figure 9 confirms a high feature survival rate across blur gaps, proving the algorithm recovers without indefinite pausing.

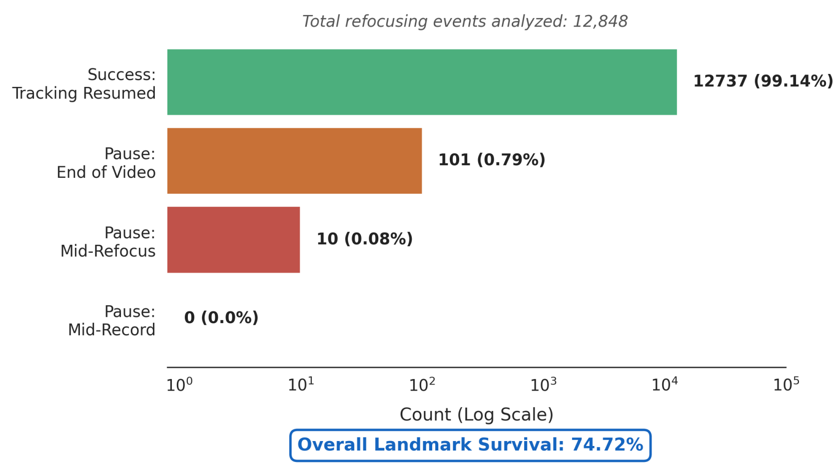


Figure 9: **Distribution of tracking recovery outcomes across 12,848 refocusing events.** The high success rate and total absence of mid-video pausing and strong landmark survival rate confirms the mechanism operates flawlessly, with non-resumed instances occurring only when the recording ends during still or blurry frames.

Hyperparameter tuning. To attain final hyperparameter settings for our model, we conduct an extensive hyperparameter tuning phase across five key configuration thresholds of the Optical Flow algorithm, as shown in Figure 10. The system's downstream classification performance, measured by Validation Accuracy, F1-Score, and AUC, was recorded for each parameter variation to identify the optimal configuration for identifying frames containing diagnostically relevant regions, which is described in Appendix, Section B.

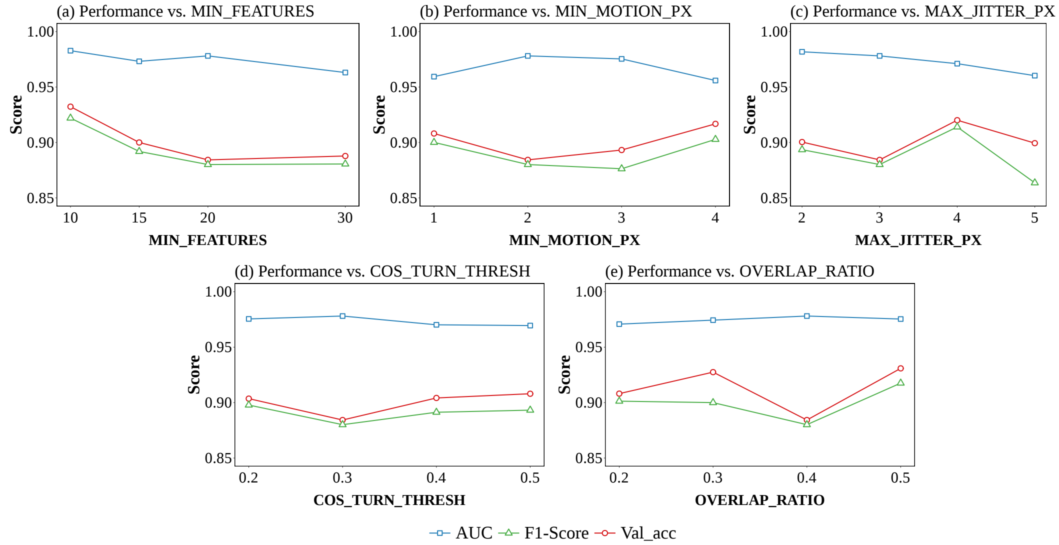


Figure 10: **Hyperparameter tuning for our proposed frame selection algorithm.** Variations in predictive performance, measured by Accuracy, F1-Score, and Area Under the ROC Curve (AUC), are recorded across different configurations of five hyperparameters for our frame sampling strategy.

H Test-Time Compute Evaluation

We evaluate the inference efficiency of PathFlow by analyzing the relationship between input video duration and processing time. Figure 11 reports the average inference time across different video length intervals.

As observed, inference time increases with video duration; however, the overall computational overhead remains consistently low. Even for long videos exceeding 5 minutes, the average processing time remains bounded (within $\sim$2 minutes), indicating that the additional cost introduced by longer inputs is minimal.

This efficiency is enabled by the architectural design of PathFlow. The optical-flow-based frame selection stage removes redundant frames early, while the prototype-based aggregation compresses variable-length sequences into a compact set of tokens. Consequently, the downstream Transformer operates on a bounded input size, effectively decoupling inference complexity from the raw number of frames.

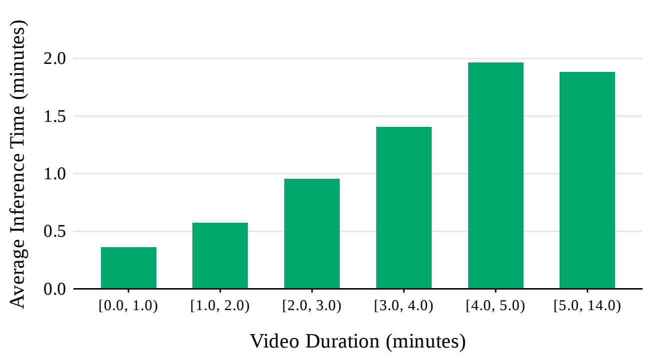


Figure 11: **Average inference time across different video duration intervals.** Overall, most videos require around 1-2 minutes to obtain a classification during the inference stage.

Importantly, the inference latency remains low and practically negligible relative to the variability in input video length. In the context of clinical deployment (Section D), where end-to-end processing time is typically 2-5 minutes per case, the model inference itself constitutes only a small fraction of the total pipeline. This demonstrates that PathFlow achieves a favorable balance between scalability and efficiency, making it suitable for real-world diagnostic workflows.

I Class-wise Error Analysis

We evaluate one-vs-rest confusion matrices on a 107-sample validation set, as shown in Figure 12 to identify misclassification drivers across the three classes:

- **Normal:** Highest overall accuracy (TP=27, TN=77, FP=1, FN=2).
- **Adenoma:** Highest False Positives (FP=4; TP=38, TN=63, FN=2). Errors are driven by its nature as a pre-cancerous intermediate, exhibiting visual overlap with both normal and malignant tissues.
- **Malignant:** Highest False Negatives (FN=4; TP=34, TN=66, FP=3). Misclassifications primarily occur when malignant indicators are subtle or lack consistency across video frames.

We attribute misclassifications to two primary error sources. Firstly, borderline anatomical characteristics between the Adenoma and Malignant stages may induce confusion for the model during its learning stage. Secondly, data artifacts obscuring the fine-grained textures needed for high-confidence boundary decisions may cause noisy data.

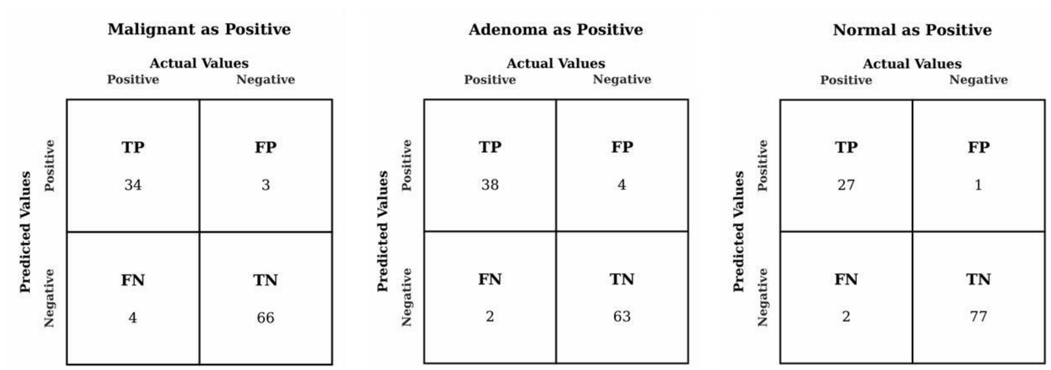


Figure 12: **Failure case analysis.** One-vs-rest binary confusion matrices for Malignant, Adenoma, and Normal classes as positive.

J Motion-Aware Adaptive Frame Selection

Algorithm Description. Algorithm 1 proceeds as follows:

1. **Initialization.** The selected set X is initialized with the first frame f_1, which serves as the initial reference index ($prev$). The motion accumulator A is set to zero, and the motion history H is initialized as an empty set.
2. **Feature Extraction.** For each incoming frame f_t, central crops f^c_{prev} and f^c_t are extracted. Shi–Tomasi keypoints P_{prev} are detected on the reference crop. If the number of features $|P_{prev}|$ falls below a threshold T_{feat}, the frame is skipped to avoid unreliable estimation.
3. **Motion Estimation.** Detected keypoints are tracked to the current crop using Lucas–Kanade optical flow. The dominant motion vector Δ_t is computed as the median displacement of all tracked points. Additionally, a jitter measure J_t is calculated as the median absolute deviation of individual point motions to quantify inconsistency.
4. **Frame Filtering.** Frames are discarded if they represent a static scene (magnitude $s_t < T_{\text{motion}}$) or if the motion is excessively unstable ($J_t > T_{\text{jitter}}$).

5. **Adaptive Scaling.** The algorithm maintains a history H of recent motion magnitudes. If the current speed s_t is significantly lower than the historical median μ (scaled by τ), an inspection factor α_{insp} is applied to adjust the spatial selection sensitivity.
6. **Motion Accumulation.** The current motion vector Δ_t is added to the accumulator A, which tracks the total spatial displacement since the last selected frame.
7. **Frame Selection.** A frame is added to X if either of two triggers is met:
   - **Spatial Coverage:** The accumulated displacement $\|A\|$ exceeds a threshold defined by the frame width W, the overlap ratio λ, and the adaptive factor α.
   - **Directional Change:** The cosine similarity between the current motion and the accumulated trajectory falls below T_{turn}, indicating a significant change in heading.
8. **Reset and Iteration.** Upon selection, the reference index is updated ($prev \leftarrow t$), the accumulator A is reset to zero, and the process continues until all frames are processed.

Algorithm 1 Motion-Aware Adaptive Frame Selection

Input: Frames $F = \{f_t\}_{t=1}^{T}$, $f_t \in \mathbb{R}^{H \times W \times 3}$
Parameters: Feature threshold T_{feat}, motion thresholds $T_{\text{motion}}, T_{\text{jitter}}$, turning threshold T_{turn}, overlap ratio λ, history length N_{hist}, speed ratio τ, inspection factor α_{insp}
Output: Selected frame set X

Initialize $X \leftarrow \{f_1\}$, $prev \leftarrow 1$
Initialize $A \leftarrow \mathbf{0}$, $H \leftarrow \emptyset$
for $t = 2$ to T **do**
    Extract central crops f^c_{prev}, f^c_t
    Detect D_p Shi-Tomasi keypoints P_{prev} in f^c_{prev}
    if $|P_{prev}| < T_{\text{feat}}$ **then**
        continue
    end if
    Track P_{prev} to f^c_t via Lucas-Kanade flow
    Compute $v_u = p^t_u - p^{prev}_u$ and $\Delta_t = \text{median}_u(v_u)$
    $s_t = \|\Delta_t\|, \quad J_t = \text{median}_u(\|v_u - \Delta_t\|)$
    if $s_t < T_{\text{motion}}$ **or** $J_t > T_{\text{jitter}}$ **then**
        continue
    end if
    Update $H \leftarrow (H \cup \{s_t\})_{[-N_{\text{hist}}:]}, \quad \mu = \text{median}(H)$
    $\alpha \leftarrow \alpha_{\text{insp}}$ **if** $s_t < \tau\mu$ **else** 1.0
    $A \leftarrow A + \Delta_t$
    $Trigger_{\text{dist}} \leftarrow \|A\| \geq \alpha W(1-\lambda)$
    $Trigger_{\text{dir}} \leftarrow \frac{\Delta_t \cdot A}{\|\Delta_t\|\|A\|} < T_{\text{turn}}$
    if $Trigger_{\text{dist}}$ **or** $Trigger_{\text{dir}}$ **then**
        $X \leftarrow X \cup \{f_t\}$
        $prev \leftarrow t, A \leftarrow \mathbf{0}$
    end if
end for
return X

K Discussion

Although PathFlow demonstrates strong performance and a clear pathway toward deployment, inference time remains an important practical consideration. In our survey conducted with pathologists with more than 10 years of clinical experience, several reported that they can complete a full manual scan and diagnostic assessment of a digital whole-slide image within approximately 1-2 minutes, which is significantly faster than the average 5-10 minutes. In contrast, the current implementation of PathFlow requires around 2-5 minutes per case, depending on video length, available computational resources, and processing time of on-side infrastructure. This gap highlights that, while PathFlow substantially reduces the workload relative to scanner-based workflows, its inference speed has not yet reached the level of expert manual review. Addressing this limitation, we plan to explore strategies such as quantization techniques and structured pruning to reduce inference time while maintaining diagnostic performance. These optimizations aim to better align system inference latency with the time constraints of routine clinical practice.

Furthermore, while MVP-CRC is currently the first and only dataset that explores microscope-based videos in pathology, we intend to create more datasets of a similar nature as MVP-CRC to further verify and enhance PathFlow's robustness in the future. We also plan to expand the MVP-CRC dataset by creating annotations for diagnostically relevant ROIs at the frame level, which can be used to perform polyp subtyping. Overall, existing computational pathology studies and datasets mainly focus on digital slides acquired from whole-slide scanners, and microscope-acquired videos remain a largely unexplored domain despite being the primary diagnostic workflow in many developing countries, as discussed in Section 1. This discrepancy creates a major gap between AI research and practical clinical deployment. Our work aims to bridge this gap by introducing a large-scale microscopic pathology video dataset and a dedicated framework for this modality. At its current stage, this work aims to establish a first-ever benchmark for microscopic video analysis in computational pathology.

It is also worth noting that although PathFlow demonstrates strong performance on the MVP-CRC dataset, the system is intended to support rather than replace expert pathologists in real-world clinical workflows. As an AI-powered framework, PathFlow may still produce incorrect predictions, particularly under unseen acquisition conditions, rare pathological patterns, or low staining and microscope quality. Therefore, PathFlow should be used responsibly as a clinical decision-support tool designed to assist pathologists by highlighting diagnostically relevant patterns and reducing workload, rather than an autonomous diagnostic system in itself. Final clinical decisions must remain under the supervision of qualified medical professionals. Moreover, careful management of medical data and adherence to institutional privacy regulations remain necessary for responsible deployment in real clinical circumstances.

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